

Edward Osei-Nyarko

edward.osei.nyarko@gmail.com | +233 205 883 590 | Accra, Ghana
github.com/nana-yaw | linkedin.com/in/edward-osei-nyarko-5789b3118 |
edwardoseidev.vercel.app

Summary

Full-stack engineer with 4+ years building scalable APIs, real-time systems, and production web applications for SaaS platforms and community organizations. Designed database architectures serving 1,000+ users and automation frameworks that cut test execution time by 50%. Built a full-stack PWA from scratch with 30+ database tables, 6-role RBAC, AI-powered analytics, and 280+ tests. Shipped a real-time event management app with Firebase multi-device sync used at a live 100+ attendee event. Experienced working remotely with US-based teams in fast-paced startup environments.

Technical Skills

Languages: PHP, TypeScript, JavaScript, Node.js, Kotlin, Java, SQL, Bash
Frontend: React 19, Next.js 16 (App Router), Tailwind CSS, Radix UI, Framer Motion
Backend: Symfony 6, Laravel, Spring Boot, Convex (real-time serverless)
Databases: PostgreSQL, MySQL, Firebase Realtime DB, Convex DB (document-relational)
Testing: Cypress, Playwright, Vitest, PHPUnit, JUnit, Stryker (mutation testing)
Cloud & DevOps: Docker, AWS, Linux, Nginx, CI/CD, Vercel
APIs & Auth: REST, GraphQL, PASETO, Magic Link (Resend), Web Push (VAPID)
AI: Google Gemini API integration
Tools: Git, Swagger/OpenAPI, JIRA, Postman, Agile/Scrum

Experience

Backend Engineer — RentPost, Inc. (Remote, US) | Jul 2023 -- Present

- Designed PostgreSQL database isolation architecture for parallel test execution using template-based cloning, reducing total test runtime by 50% with 2-4 concurrent workers
- Built cross-repository orchestration layer decoupling frontend and backend tests through CLI interfaces, establishing clean service boundaries and improving maintainability
- Developed PASETO token-based authentication system for test automation, enabling secure stateless API access without session overhead
- Engineered parallel test runner with timing persistence and data-driven load balancing, maintaining 92% pass rate across 24 automated tests
- Refactored infrastructure scripts following YAGNI principles, reducing codebase by 61% while preserving functionality
- Developed and optimized SaaS APIs using PHP/Symfony with PSR-12 compliance; wrote comprehensive technical documentation
- Implemented interactive onboarding tour with driver.js, improving user adoption and reducing support tickets

Software Engineer — Turntabl Ghana Ltd. (Accra, Ghana) | 2021 -- 2023

- Built News Aggregator REST API using Spring Boot and Kotlin, scaling from 100 to 1,000+ users after company website integration
 - Optimized API performance for internal systems; wrote technical documentation for engineer onboarding
 - Mentored junior engineers on best practices, fostering team growth and improving code quality
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Projects

EWC Care App — Full-Stack Pastoral Care Management PWA

Stack: Next.js 16, React 19, TypeScript, Convex, Tailwind CSS, Google Gemini AI

Links: [ewcareapp.online](#) | [Portfolio](#)

- Designed and built a production PWA serving 5+ communities with real-time data synchronization across 30+ database tables and 92 indexes
- Implemented 6-role RBAC + ABAC permission system (super admin to viewer) with PII field projection, scope-based access control, and enterprise audit logging with SHA-256 integrity chains
- Built AI-powered pastoral analytics using Google Gemini for member engagement insights and care recommendations

- Developed GPS-geofenced attendance check-in with Haversine validation and spoofing detection
- Created automated Circle Sunday algorithm for periodic member reshuffling across community groups
- Wrote 280+ tests (unit, component, security, E2E) using Vitest, Cypress, and Playwright with mutation testing via Stryker
- Implemented magic-link authentication via Resend, web push notifications (VAPID), QR code attendance, and offline-first PWA with service worker

EWC Teams Bootcamp — Real-Time Event Management App

Stack: Next.js 15, React 19, TypeScript, Firebase Realtime DB, Tailwind CSS, Playwright

Links: teambootcampapp.vercel.app | [Portfolio](#)

- Built a real-time volunteer management and scoring system for a 100+ attendee church bootcamp with multi-device sync (projector, referees, attendees)
 - Implemented atomic Firebase transactions for concurrent team assignment, preventing race conditions during simultaneous signups
 - Built live leaderboard with FLIP-based animations, admin scoring portal, role-based access (3 tiers with PIN auth), and Excel data export
 - Wrote Playwright E2E test suite with Firebase emulator fixtures for local testing
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Education

B.Sc. Physics — University of Cape Coast, Ghana (2015)